

Concavity, Curve Sketching, and Optimization

ANSWER KEY



Increasing/decreasing and the first derivative test

- 1 For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. $f' > 0$ on $(-\infty, 1)$, $f' < 0$ on $(1, 3)$, $f' > 0$ on $(3, \infty)$. So f is increasing on $(-\infty, 1) \cup (3, \infty)$, decreasing on $(1, 3)$. $x = 1$ is a local max (sign $+\rightarrow -$); $x = 3$ is a local min (sign $-\rightarrow +$).

Concavity and the second derivative test

- 2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f'(x) = 4x^3 - 12x^2$; $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ on $(-\infty, 0)$, $f'' < 0$ on $(0, 2)$, $f'' > 0$ on $(2, \infty)$. Concave up on $(-\infty, 0) \cup (2, \infty)$, concave down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$ (both sign changes).

- 3 Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.

$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$. $f''(x) = 6x$. $f''(1) = 6 > 0$: local min at $x = 1$. $f''(-1) = -6 < 0$: local max at $x = -1$.

Curve sketching

- 4 For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.

$f'(x) = 3x^2 - 6x = 3x(x-2)$: critical points $x = 0, 2$. $f''(x) = 6x - 6$: zero at $x = 1$. Sign chart: $f' > 0$ on $(-\infty, 0)$, $f' < 0$ on $(0, 2)$, $f' > 0$ on $(2, \infty)$ — local max at $x = 0$ ($f(0) = 0$), local min at $x = 2$ ($f(2) = -4$). $f'' < 0$ on $(-\infty, 1)$ (concave down), $f'' > 0$ on $(1, \infty)$ (concave up) — inflection point at $x = 1$ ($f(1) = -2$).

Optimization problems

- 5 A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

Let x = side parallel to river, y = each perpendicular side. Constraint: $x + 2y = 200$, so $x = 200 - 2y$. Area $A = xy = (200 - 2y)y = 200y - 2y^2$. $A'(y) = 200 - 4y = 0 \Rightarrow y = 50$. $x = 200 - 100 = 100$. Maximum area occurs at $x = 100$ m, $y = 50$ m (verified max since $A'' = -4 < 0$).

- 6 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.

$$V(x) = x(12 - 2x)^2 \text{ for } 0 < x < 6.$$

$$V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)[(12 - 2x) - 4x] = (12 - 2x)(12 - 6x). \text{ Setting } V' = 0: x = 6 \text{ (rejected, gives } V = 0) \text{ or } x = 2. V(2) = 2(8)^2 = 128 \text{ in}^3, \text{ the maximum.}$$

Optimization: minimizing cost or distance

- 7 Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

$$D(x) = x^2 + (2x - 3)^2 = x^2 + 4x^2 - 12x + 9 = 5x^2 - 12x + 9. D'(x) = 10x - 12 = 0 \Rightarrow x = \frac{6}{5}. \\ y = 2\left(\frac{6}{5}\right) - 3 = \frac{12}{5} - 3 = -\frac{3}{5}. \text{ Closest point: } \left(\frac{6}{5}, -\frac{3}{5}\right) \text{ (confirmed min since } D'' = 10 > 0).$$

- 8 A closed cylindrical can must have volume $16\pi \text{ cm}^3$. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes surface area.

$$\text{From } V: h = \frac{16}{r^2}. S(r) = 2\pi r^2 + 2\pi r \cdot \frac{16}{r^2} = 2\pi r^2 + \frac{32\pi}{r}. S'(r) = 4\pi r - \frac{32\pi}{r^2} = 0 \Rightarrow 4\pi r = \frac{32\pi}{r^2} \Rightarrow r^3 = 8 \Rightarrow r = 2 \text{ cm.}$$

Points of inflection and graph behavior

- 9 Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

$f'(x) = 3x^2 \geq 0$ everywhere, so f is never decreasing — $x = 0$ is not a local extremum. However $f''(x) = 6x$ changes sign at $x = 0$ (from concave down to concave up), so $x = 0$ IS an inflection point. This shows concavity change and extrema are independent features.

Visualizing extrema and concavity together

- 10 The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.

The graph rises to a peak at $x = 0$ ($f = 0$), then falls, changing its bend from concave-down to concave-up right at $x = 1$ ($f = -2$, the inflection point), before reaching a valley at $x = 2$ ($f = -4$) and rising again — visually confirming all three features found algebraically.

A capstone optimization problem

- 11 A rectangular box with a square base and no top must have volume 500 cm^3 . Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.
- $$S(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}. S'(x) = 2x - \frac{2000}{x^2} = 0 \Rightarrow 2x^3 = 2000 \Rightarrow x^3 = 1000 \Rightarrow x = 10 \text{ cm}.$$

Second derivative test failures

- 12 For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.

$f'(x) = 4x^3 = 0$ gives $x = 0$. $f''(x) = 12x^2$; $f''(0) = 0$, so the Second Derivative Test gives no information. Using the First Derivative Test: $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$, so $x = 0$ is a local (and in fact absolute) minimum.

A capstone curve-sketching problem

- 13 For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x - 2)(x + 1)$: critical points $x = -1, 0, 2$.

$f''(x) = 36x^2 - 24x - 24$. $f''(-1) = 36 + 24 - 24 = 36 > 0$: local min. $f''(0) = -24 < 0$: local max.

$f''(2) = 144 - 48 - 24 = 72 > 0$: local min.

Multiple choice — easy 15

- 14 Classify the critical point $x = -1$ of $f(x) = x^3 + 6x^2 + 9x$, using the Second Derivative Test.

- a. a local maximum
- b. cannot be determined without more information
- c. a local minimum ✓
- d. an inflection point

$f''(x) = 6x + 12$, so $f''(-1) = -6 + 12 = 6$, which is positive: $x = -1$ is a local minimum.

15 Classify the critical point $x = -1$ of $f(x) = x^3 - 6x^2 - 15x$, using the Second Derivative Test.

- a. a local maximum ✓
- b. a local minimum
- c. cannot be determined without more information
- d. an inflection point

$f''(x) = 6x - 12$, so $f''(-1) = -6 - 12 = -18$, which is negative: $x = -1$ is a local maximum.

16 Classify the critical point $x = 1$ of $f(x) = x^3 + 3x^2 - 9x$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. a local minimum ✓
- c. an inflection point
- d. a local maximum

$f''(x) = 6x + 6$, so $f''(1) = 6 + 6 = 12$, which is positive: $x = 1$ is a local minimum.

17 Classify the critical point $x = 5$ of $f(x) = x^3 - 3x^2 - 45x$, using the Second Derivative Test.

- a. a local maximum
- b. an inflection point
- c. a local minimum ✓
- d. cannot be determined without more information

$f''(x) = 6x - 6$, so $f''(5) = 30 - 6 = 24$, which is positive: $x = 5$ is a local minimum.

18 Classify the critical point $x = -6$ of $f(x) = x^3 + 12x^2 + 36x$, using the Second Derivative Test.

- a. a local minimum
- b. a local maximum ✓
- c. an inflection point
- d. cannot be determined without more information

$f''(x) = 6x + 24$, so $f''(-6) = -36 + 24 = -12$, which is negative: $x = -6$ is a local maximum.

19 Classify the critical point $x = 0$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. a local minimum ✓**
- c. a local maximum
- d. an inflection point

$f''(x) = 6x + 18$, so $f''(0) = 0 + 18 = 18$, which is positive: $x = 0$ is a local minimum.

20 Classify the critical point $x = 0$ of $f(x) = x^3 - 9x^2$, using the Second Derivative Test.

- a. a local maximum ✓**
- b. an inflection point
- c. cannot be determined without more information
- d. a local minimum

$f''(x) = 6x - 18$, so $f''(0) = 0 - 18 = -18$, which is negative: $x = 0$ is a local maximum.

21 Classify the critical point $x = 5$ of $f(x) = x^3 - 6x^2 - 15x$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. an inflection point
- c. a local maximum
- d. a local minimum ✓**

$f''(x) = 6x - 12$, so $f''(5) = 30 - 12 = 18$, which is positive: $x = 5$ is a local minimum.

22 Classify the critical point $x = 6$ of $f(x) = x^3 - 3x^2 - 72x$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. an inflection point
- c. a local maximum
- d. **a local minimum ✓**

$f''(x) = 6x - 6$, so $f''(6) = 36 - 6 = 30$, which is positive: $x = 6$ is a local minimum.

23 Classify the critical point $x = 4$ of $f(x) = x^3 - 15x^2 + 72x$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. a local minimum
- c. **a local maximum ✓**
- d. an inflection point

$f''(x) = 6x - 30$, so $f''(4) = 24 - 30 = -6$, which is negative: $x = 4$ is a local maximum.

24 Classify the critical point $x = 4$ of $f(x) = x^3 - 9x^2 + 24x$, using the Second Derivative Test.

- a. a local maximum
- b. an inflection point
- c. **a local minimum ✓**
- d. cannot be determined without more information

$f''(x) = 6x - 18$, so $f''(4) = 24 - 18 = 6$, which is positive: $x = 4$ is a local minimum.

25 Classify the critical point $x = 6$ of $f(x) = x^3 - 15x^2 + 72x$, using the Second Derivative Test.

- a. a local maximum
- b. cannot be determined without more information
- c. **a local minimum ✓**
- d. an inflection point

$f''(x) = 6x - 30$, so $f''(6) = 36 - 30 = 6$, which is positive: $x = 6$ is a local minimum.

26 Classify the critical point $x = -6$ of $f(x) = x^3 + 9x^2$, using the Second Derivative Test.

- a. an inflection point
- b. a local minimum
- c. a local maximum ✓**
- d. cannot be determined without more information

$f''(x) = 6x + 18$, so $f''(-6) = -36 + 18 = -18$, which is negative: $x = -6$ is a local maximum.

27 Classify the critical point $x = -2$ of $f(x) = x^3 + 3x^2$, using the Second Derivative Test.

- a. cannot be determined without more information
- b. an inflection point
- c. a local maximum ✓**
- d. a local minimum

$f''(x) = 6x + 6$, so $f''(-2) = -12 + 6 = -6$, which is negative: $x = -2$ is a local maximum.

28 Classify the critical point $x = -5$ of $f(x) = x^3 + 3x^2 - 45x$, using the Second Derivative Test.

- a. an inflection point
- b. a local maximum ✓**
- c. cannot be determined without more information
- d. a local minimum

$f''(x) = 6x + 6$, so $f''(-5) = -30 + 6 = -24$, which is negative: $x = -5$ is a local maximum.

Multiple choice — medium 15

29 Find the nonzero inflection point of $f(x) = x^4 + 6x^3$.

a. -3 ✓

b. -6

c. -2

d. 6

$f''(x) = 12x^2 + 36x = 6x(2x + 6)$. Zero at $x = 0$ and $x = -3$; the nonzero inflection point is $x = -3$.

30 Find the nonzero inflection point of $f(x) = x^4 + 14x^3$.

a. 14

b. -6

c. -14

d. -7 ✓

$f''(x) = 12x^2 + 84x = 6x(2x + 14)$. Zero at $x = 0$ and $x = -7$; the nonzero inflection point is $x = -7$.

31 Find the nonzero inflection point of $f(x) = x^4 + 10x^3$.

a. -4

b. -5 ✓

c. 10

d. -10

$f''(x) = 12x^2 + 60x = 6x(2x + 10)$. Zero at $x = 0$ and $x = -5$; the nonzero inflection point is $x = -5$.

32 Find the nonzero inflection point of $f(x) = x^4 - 2x^3$.

a. -2

b. 2

c. 4

d. 1 ✓

$f''(x) = 12x^2 - 12x = 6x(2x - 2)$. Zero at $x = 0$ and $x = 1$; the nonzero inflection point is $x = 1$.

33 Find the nonzero inflection point of $f(x) = x^4 + 4x^3$.

- a. -1
- b. -4
- c. -2 ✓
- d. 4

$f''(x) = 12x^2 + 24x = 6x(2x + 4)$. Zero at $x = 0$ and $x = -2$; the nonzero inflection point is $x = -2$.

34 Find the nonzero inflection point of $f(x) = x^4 - 14x^3$.

- a. 8
- b. 14
- c. 7 ✓
- d. -14

$f''(x) = 12x^2 - 84x = 6x(2x - 14)$. Zero at $x = 0$ and $x = 7$; the nonzero inflection point is $x = 7$.

35 Find the nonzero inflection point of $f(x) = x^4 + 12x^3$.

- a. -5
- b. -12
- c. 12
- d. -6 ✓

$f''(x) = 12x^2 + 72x = 6x(2x + 12)$. Zero at $x = 0$ and $x = -6$; the nonzero inflection point is $x = -6$.

36 Find the nonzero inflection point of $f(x) = x^4 - 12x^3$.

a. 7

b. 6 ✓

c. -12

d. 12

$f''(x) = 12x^2 - 72x = 6x(2x - 12)$. Zero at $x = 0$ and $x = 6$; the nonzero inflection point is $x = 6$.

37 Find the nonzero inflection point of $f(x) = x^4 + 16x^3$.

a. -7

b. -16

c. -8 ✓

d. 16

$f''(x) = 12x^2 + 96x = 6x(2x + 16)$. Zero at $x = 0$ and $x = -8$; the nonzero inflection point is $x = -8$.

38 Find the nonzero inflection point of $f(x) = x^4 - 8x^3$.

a. 5

b. 8

c. -8

d. 4 ✓

$f''(x) = 12x^2 - 48x = 6x(2x - 8)$. Zero at $x = 0$ and $x = 4$; the nonzero inflection point is $x = 4$.

39 Find the nonzero inflection point of $f(x) = x^4 - 4x^3$.

a. 4

b. -4

c. 3

d. 2 ✓

$f''(x) = 12x^2 - 24x = 6x(2x - 4)$. Zero at $x = 0$ and $x = 2$; the nonzero inflection point is $x = 2$.

40 Find the nonzero inflection point of $f(x) = x^4 - 10x^3$.

a. 5 ✓

b. 6

c. -10

d. 10

$f''(x) = 12x^2 - 60x = 6x(2x - 10)$. Zero at $x = 0$ and $x = 5$; the nonzero inflection point is $x = 5$.

41 Find the nonzero inflection point of $f(x) = x^4 + 2x^3$.

a. 2

b. -1 ✓

c. -2

d. -3

$f''(x) = 12x^2 + 12x = 6x(2x + 2)$. Zero at $x = 0$ and $x = -1$; the nonzero inflection point is $x = -1$.

42 Find the nonzero inflection point of $f(x) = x^4 - 16x^3$.

a. 16

b. 8 ✓

c. -16

d. 9

$f''(x) = 12x^2 - 96x = 6x(2x - 16)$. Zero at $x = 0$ and $x = 8$; the nonzero inflection point is $x = 8$.

43 Find the nonzero inflection point of $f(x) = x^4 + 8x^3$.

- a. -3
- b. -4 ✓
- c. 8
- d. -8

$f''(x) = 12x^2 + 48x = 6x(2x + 8)$. Zero at $x = 0$ and $x = -4$; the nonzero inflection point is $x = -4$.

Multiple choice — hard 15

44 A farmer has 100 m of fencing to enclose a rectangular field with $x + 2y = 100$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

- a. 26
- b. 100
- c. 25 ✓
- d. 50

$A(y) = (100 - 2y)y = 100y - 2y^2$. $A'(y) = 100 - 4y = 0 \Rightarrow y = \frac{100}{4} = 25$ m.

45 A farmer has 72 m of fencing to enclose a rectangular field with $x + 2y = 72$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

- a. 72
- b. 36
- c. 19
- d. 18 ✓

$A(y) = (72 - 2y)y = 72y - 2y^2$. $A'(y) = 72 - 4y = 0 \Rightarrow y = \frac{72}{4} = 18$ m.

- 46 A farmer has 148 m of fencing to enclose a rectangular field with $x + 2y = 148$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 37 ✓

b. 148

c. 74

d. 38

$$A(y) = (148 - 2y)y = 148y - 2y^2. A'(y) = 148 - 4y = 0 \Rightarrow y = \frac{148}{4} = 37 \text{ m.}$$

- 47 A farmer has 108 m of fencing to enclose a rectangular field with $x + 2y = 108$ (one side of length x needs no fence, the other 2 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 28

b. 108

c. 27 ✓

d. 54

$$A(y) = (108 - 2y)y = 108y - 2y^2. A'(y) = 108 - 4y = 0 \Rightarrow y = \frac{108}{4} = 27 \text{ m.}$$

- 48 A farmer has 168 m of fencing to enclose a rectangular field with $x + 3y = 168$ (one side of length x needs no fence, the other 3 side(s) of length y each do). Find y that maximizes the enclosed area $A = xy$.

a. 168

b. 29

c. 28 ✓

d. 56

$$A(y) = (168 - 3y)y = 168y - 3y^2. A'(y) = 168 - 6y = 0 \Rightarrow y = \frac{168}{6} = 28 \text{ m.}$$

- 49 An open-top box is made from a 36 in by 36 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(36 - 2x)^2$.

a. 6 ✓

b. 18

c. 9

d. 7

$V'(x) = (36 - 2x)^2 + x \cdot 2(36 - 2x)(-2) = (36 - 2x)[(36 - 2x) - 4x] = (36 - 2x)(36 - 6x)$. Setting $V' = 0$: $x = \frac{36}{2}$ (rejected, gives $V = 0$) or $x = \frac{36}{6} = 6$, the maximum.

- 50 An open-top box is made from a 18 in by 18 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(18 - 2x)^2$.

a. 3 ✓

b. 6

c. 9

d. 4

$V'(x) = (18 - 2x)^2 + x \cdot 2(18 - 2x)(-2) = (18 - 2x)[(18 - 2x) - 4x] = (18 - 2x)(18 - 6x)$. Setting $V' = 0$: $x = \frac{18}{2}$ (rejected, gives $V = 0$) or $x = \frac{18}{6} = 3$, the maximum.

- 51 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(12 - 2x)^2$.

a. 4

b. 2 ✓

c. 3

d. 6

$V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)[(12 - 2x) - 4x] = (12 - 2x)(12 - 6x)$. Setting $V' = 0$: $x = \frac{12}{2}$ (rejected, gives $V = 0$) or $x = \frac{12}{6} = 2$, the maximum.

- 52 An open-top box is made from a 24 in by 24 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(24 - 2x)^2$.

a. 4 ✓

b. 5

c. 12

d. 6

$V'(x) = (24 - 2x)^2 + x \cdot 2(24 - 2x)(-2) = (24 - 2x)[(24 - 2x) - 4x] = (24 - 2x)(24 - 6x)$. Setting $V' = 0$: $x = \frac{24}{2}$ (rejected, gives $V = 0$) or $x = \frac{24}{6} = 4$, the maximum.

- 53 An open-top box is made from a 48 in by 48 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume $V(x) = x(48 - 2x)^2$.

a. 9

b. 8 ✓

c. 24

d. 12

$V'(x) = (48 - 2x)^2 + x \cdot 2(48 - 2x)(-2) = (48 - 2x)[(48 - 2x) - 4x] = (48 - 2x)(48 - 6x)$. Setting $V' = 0$: $x = \frac{48}{2}$ (rejected, gives $V = 0$) or $x = \frac{48}{6} = 8$, the maximum.

- 54 Find the x -coordinate of the point on the line $y = -5x - 4$ closest to the origin, by minimizing $D = x^2 + y^2$.

a. $-\frac{4}{5}$

b. $\frac{2}{13}$

c. $-\frac{10}{13}$ ✓

d. $\frac{10}{13}$

$D(x) = x^2 + (-5x - 4)^2 = (26)x^2 + 40x + 16$. $D'(x) = 52x + 40 = 0 \Rightarrow x = \frac{-20}{26} = -\frac{10}{13}$.

55 Find the x-coordinate of the point on the line $y = 4x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

- a. $-\frac{5}{17}$
- b. $-\frac{5}{4}$
- c. $-\frac{20}{17}$ ✓
- d. $\frac{20}{17}$

$$D(x) = x^2 + (4x + 5)^2 = (17)x^2 + 40x + 25. D'(x) = 34x + 40 = 0 \Rightarrow x = \frac{-20}{17} = -\frac{20}{17}.$$

56 Find the x-coordinate of the point on the line $y = 5x - 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

- a. 1
- b. $\frac{25}{26}$ ✓
- c. $-\frac{25}{26}$
- d. $\frac{5}{26}$

$$D(x) = x^2 + (5x - 5)^2 = (26)x^2 - 50x + 25. D'(x) = 52x - 50 = 0 \Rightarrow x = \frac{25}{26} = \frac{25}{26}.$$

57 Find the x-coordinate of the point on the line $y = x + 3$ closest to the origin, by minimizing $D = x^2 + y^2$.

- a. $-\frac{3}{2}$ ✓
- b. -3
- c. $\frac{3}{2}$
- d. $-\frac{1}{2}$

$$D(x) = x^2 + (x + 3)^2 = (2)x^2 + 6x + 9. D'(x) = 4x + 6 = 0 \Rightarrow x = \frac{-3}{2} = -\frac{3}{2}.$$

58 Find the x-coordinate of the point on the line $y = x + 5$ closest to the origin, by minimizing $D = x^2 + y^2$.

- a. -5
- b. $-\frac{5}{2}$ ✓
- c. -3
- d. $\frac{5}{2}$

$$D(x) = x^2 + (x + 5)^2 = (2)x^2 + 10x + 25. D'(x) = 4x + 10 = 0 \Rightarrow x = \frac{-5}{2} = -\frac{5}{2}.$$